

Mohammed Yaqoob

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EXPERIENCE

Software Engineering Intern

Jan 2025 – May 2025

Stalwart.ae · Dubai, UAE (Remote)

- Delivered production features across a React + Node.js stack, owning the full cycle from spec to live deployment in an agile sprint cadence.
- Integrated third-party payment and logistics APIs with retry queues and webhook signature validation, reducing failed transaction incidents by ~40%.
- Migrated untyped REST route handlers to strict Zod-validated schemas, eliminating a class of runtime input errors at the API boundary before they reached the database.
- Established the team's first test coverage on financial flows using Jest, covering critical payment paths end-to-end.

PROJECTS

Zebra Search | *TypeScript, Node.js, SQLite Edge, React, Cloudflare Workers* | Live | GitHub Feb – Apr 2026

- Built a complete search engine from first principles — strict-mode crawler, tokenized indexer, multi-factor ranking pipeline, and query UI — with zero off-the-shelf search frameworks.
- Crawler processed 50,000+ pages with enforced quality gates (minimum content length, keyword density thresholds, duplicate URL fingerprinting) to admit only high-signal documents into the index.
- Designed a hybrid retrieval model: BM25-style lexical scoring combined with embedding-based semantic similarity, weighted by freshness decay and domain authority signals.
- Query engine backed by Cloudflare D1 (edge SQLite); globally distributed deployment achieves sub-1s end-to-end query latency with no cold-start penalty.

Perplx | *TypeScript, Next.js, Bun, Express, Turborepo, SSE* | Live | GitHub Sept – Oct 2025

- Built an AI answer engine with a fully observable inference pipeline: query intent classification → web retrieval → cross-encoder reranking → LLM response generation with citation grounding.
- Implemented a custom SSE transport layer (no third-party streaming libraries) that simultaneously delivers token-streamed responses and broadcasts live execution status per pipeline stage.
- Three-panel UI built from scratch: live pipeline stage visualizer, per-source citation confidence scoring, and an interactive knowledge graph rendered client-side without charting dependencies.
- Backend runs on Bun for native TypeScript execution without a transpile step; Express handles routing with typed middleware chain end-to-end.

Sketch.io | *React, TanStack Query, Node.js, WebSockets* | Live | GitHub Nov 2025 – Jan 2026

- Built a real-time multiplayer whiteboard supporting simultaneous canvas collaboration across N users with low-latency event propagation over WebSockets.
- Designed a custom event-sync protocol to handle concurrent draw operations from multiple clients — resolving conflicts deterministically without visible artifacts or canvas drift.
- Room and session management isolates each drawing session into independent canvas state; no cross-room bleed even under concurrent connection load.

TECHNICAL SKILLS

Languages: TypeScript, Python, Java, SQL, HTML/CSS

Frameworks & Runtimes: React, Next.js, Node.js, Express.js, FastAPI, Bun

Databases: PostgreSQL, MongoDB, Redis, MySQL, SQLite (Cloudflare D1)

Tools: Docker, Git, Turborepo, Cloudflare Workers, Prisma, Zod, WebSockets, SSE, Jest

AI / ML: LLM integration (OpenAI, Anthropic), embedding retrieval, RAG pipelines, vector scoring

EDUCATION

Shadan College of Engineering and Technology

Bachelor of Technology in Computer Science & Engineering

Hyderabad, T.S.

Nov 2022 – May 2026

ACHIEVEMENTS

- Solved 325+ DSA problems on LeetCode and HackerRank.
- Participated in 14+ contests, improving rating by 65%.